

Black Friday Dataset - SQL Checks

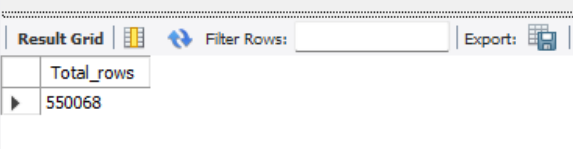
Amisha | MySQL Workbench | Data validation before Power BI

Before jumping into Power BI, I ran a few basic SQL checks on the sales table to make sure the data loaded properly - total rows, a quick spot check of random rows, and how many unique customers and products are actually in the dataset.

1. Total rows

Just confirming how many rows actually made it into the table after import.

```
3 #1. Total rows
4 • select count(*) as Total_rows from sales;
5
```



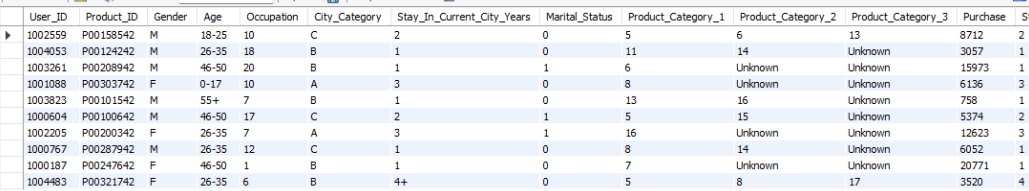
Total_rows
550068

550,068 rows - matches the original file, so nothing got dropped.

2. Spot check a few random rows

Pulled 10 random rows just to eyeball the columns and make sure everything looks right (also wanted to see how Product_Category_2 / 3 show up when a product doesn't have one - shows as 'Unknown').

```
6 # 2.Spot-check random rows
7 • SELECT * FROM sales ORDER BY RAND() LIMIT 10;
8
```



User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category_1	Product_Category_2	Product_Category_3	Purchase	St
1002559	P00158542	M	18-25	10	C	2	0	5	6	13	8712	2
1004053	P00124242	M	26-35	18	B	1	0	11	14	Unknown	3057	1
1003261	P00208942	M	46-50	20	B	1	1	6	Unknown	Unknown	15973	1
1001088	P00303742	F	0-17	10	A	3	0	8	Unknown	Unknown	6136	3
1003823	P00101542	M	55+	7	B	1	0	13	16	Unknown	758	1
1000604	P00100642	M	46-50	17	C	2	1	5	15	Unknown	5374	2
1002205	P00200342	F	26-35	7	A	3	1	16	Unknown	Unknown	12623	3
1000767	P00287942	M	26-35	12	C	1	0	8	14	Unknown	6052	1
1000187	P00247642	F	46-50	1	B	1	0	7	Unknown	Unknown	20771	1
1004483	P00321742	F	26-35	6	B	4+	0	5	8	17	3520	4

Looks fine - columns, data types, and the Unknown values are all as expected.

3. How many unique customers?

Used COUNT(DISTINCT user_id) to see how many actual people are behind the 550K rows.

```
9 # 3. distinct customers
10 • select count(distinct user_id) unique_customers from sales;
```



unique_customers
5891

5,891 unique customers. That means each person shows up an average of ~93 times in the data (makes sense since each row is one item bought, not one visit).

4. How many unique products?

Same idea, but for Product_ID.

```
12 # 4. distinct products
13 • select count(distinct product_id) unique_products from sales;
```

unique_products
3631

3,631 unique products across the dataset.

5. Quick null check

Ran counts on User_ID, Product_ID, and Purchase together against the total row count - if any of these had nulls, the count would be lower than 550,068.

```
15 • SELECT
16     COUNT(*) AS total_rows,
17     COUNT(User_ID) AS user_ids,
18     COUNT(Product_ID) AS product_ids,
19     COUNT(Purchase) AS purchases
20 FROM sales;
```

total_rows	user_ids	product_ids	purchases
550068	550068	550068	550068

All four numbers match exactly, so no missing values in these columns. Good to move on to Python/EDA and the Power BI dashboard.

Quick summary: 550,068 rows | 5,891 unique customers | 3,631 unique products | no missing values in User_ID, Product_ID or Purchase.